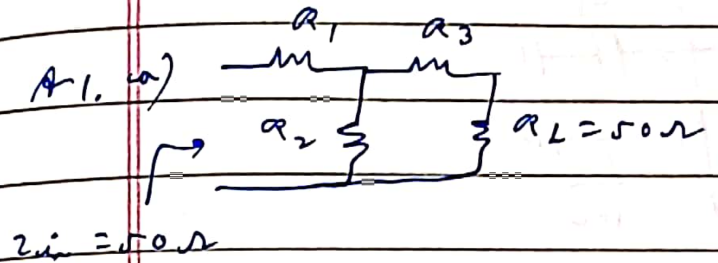
RFIC

24.3.26

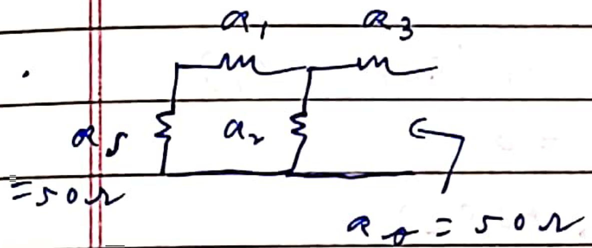
Assignment - 1

A1. a)



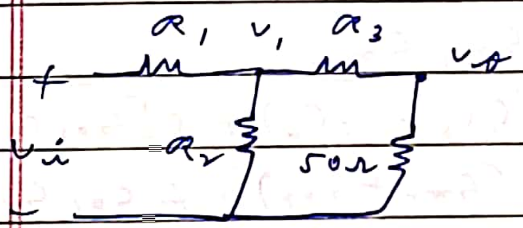
$$\therefore 50 = R_1 + (R_2 \,||\, R_3 + 50)$$

$$\therefore (50 - R_1)(R_2 + R_3 + 50) = R_2 \times (R_3 + 50) \quad -(1)$$



$$\therefore 50 = R_3 + (R_2 \,||\, R_1 + 50)$$

$$(50 - R_3)(R_2 + R_1 + 50) = R_2 \times (R_1 + 50) \quad -(2)$$



$$\frac{v_i - v_1}{R_1} = \frac{v_i}{50} \quad (\because R_{in} = 50)$$

$$\therefore v_1 = R_1 v_i \left(\frac{1}{R_1} - \frac{1}{50}\right)$$

$$\frac{v_1}{R_3 + 50} = \frac{v_o}{50} \Rightarrow \left(\frac{R_3 + 50}{50}\right) v_o = R_1 v_i \left(\frac{1}{R_1} - \frac{1}{50}\right)$$

$$\therefore \frac{v_o}{v_i} = \frac{50 R_1}{R_3 + 50}\left(\frac{1}{R_1} - \frac{1}{50}\right) = -6\,dB = \frac{1}{2} \quad -(3)$$

$$\therefore R_3 + 50 = 100 - 2R_1$$

$$R_3 = 50 - 2R_1 \quad -(4)$$

- Substituting (4) in (1) and (2):-

$$(50 - R_1)(R_2 + 100 - 2R_1) = R_2 \times (100 - 2R_1)$$

$$\therefore 2R_1^2 - 200R_1 + R_1 R_2 - 50R_2 + 5000 \quad -(5)$$

$$2R_1(R_1+R_2+50) = R_2(R_1+50)$$

$$\therefore 2R_1^2 + R_1R_2 + 100R_1 - 50R_2 = 0 \quad - (6)$$

- (5) − (6) :- $-300R_1 + 5000 = 0$

$$\therefore R_1 = \frac{50}{3}\,\Omega$$

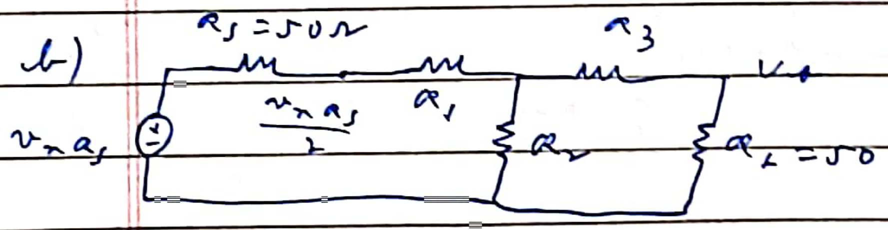
- $R_3 = 50 - \frac{100}{3} = \frac{50}{3}\,\Omega$

- From (6) :- $R_2(50-R_1) = 2R_1(R_1+50)$

$$R_2 \times \frac{100}{3} = \frac{100}{3} \times \frac{200}{3}$$

$$\therefore R_2 = \frac{200}{3}\,\Omega$$

b)



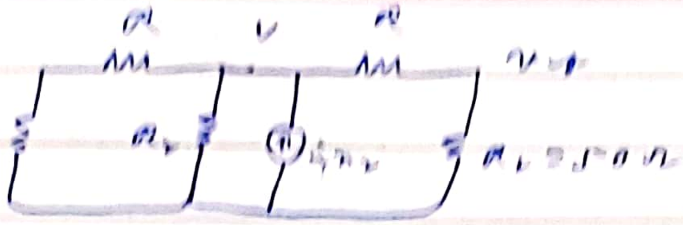
$$\frac{V_{nR_s}}{2} = 2V_o \Rightarrow V_o = \frac{V_{nR_s}}{4}$$

$$\therefore S_{V_{no}} = \frac{KTR_s}{4} = \frac{50KT}{4}$$

c) R_1 :- Same as (5) at V_{nR_s}

$$\therefore S_{V_{nR_1}} = \frac{KTR_1}{4} = \frac{50KT}{12}$$

R_2 :-



$i_{n_2} = \frac{V}{R_2} + \frac{2V}{R+50}$ $\Big\{$ $\frac{V}{R+50} = \frac{V_0}{50} \Rightarrow V = \frac{V_0(R+50)}{50}$

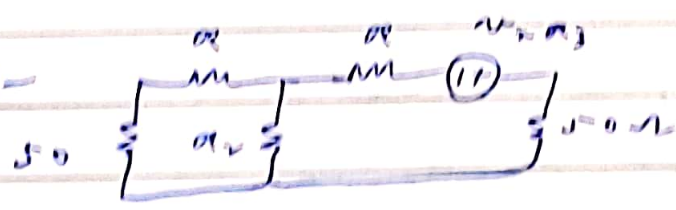
$\therefore i_{n_2} = \frac{V_0(R+50)}{50}\left(\frac{1}{R_2} + \frac{2}{R+50}\right)$

$\therefore V_0 = \frac{50 R_2 i_{n_2}}{R+50+2R_2} = \frac{50\times\frac{200}{3}\times i_{n_2}}{\frac{400+150+50}{3}}$

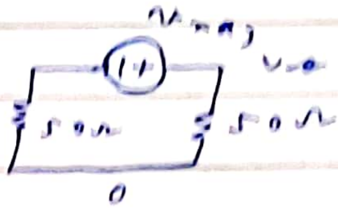
$V_0 = \frac{50\times200}{600} i_{n_2} = \frac{50}{3} i_{n_2}$

$\therefore S_{V_{n,0_2}} = \frac{2500}{9}\times\frac{4kT}{R_2} = \frac{50}{3}kT$

R_3 :-



=



$\therefore V_0 = \frac{V_{n,R_3}}{2} \Rightarrow S_{V_{n,0_3}} = kT R_3$

$S_{V_{n,0_3}} = \frac{50}{3}kT$

d) $F = 1 + \frac{kT\left(\frac{50}{12} + \frac{50}{3} + \frac{50}{3}\right)}{kT\times\frac{50}{4}} = 1 + \frac{1}{3} + \frac{8}{3}$

$= 1 + 3 = 4$

$\therefore NF = 10\log_{10}(4) = 6.02$ dB

(Q 2. a)

At $s \to 0$, $i = \dfrac{v_s}{2R+100}$

$$v_o = 50i = \frac{50 v_s}{2R+100} \Rightarrow \frac{v_o}{v_s} = \frac{25}{R+50}$$

-12 dB $\Rightarrow$ 4 times attenuation

$$\therefore \frac{1}{4} = \frac{25}{R+50} \Rightarrow R = 50$$

- $$\frac{v - v_s}{R+50} + v sC + \frac{v}{R+50} = 0 \quad \Big\{ \frac{v}{R+50} = \frac{v_o}{50}$$

$$\therefore \frac{v_s}{R+50} = \frac{(R+50)}{50} v_o \left(sC + \frac{2}{R+50} \right)$$

$$\therefore \frac{v_o}{v_s} = \left(\frac{50}{R+50} \right) \times \frac{1}{2 + sC(R+50)}$$

$$\frac{v_o}{v_s} = H(s) = \frac{1}{4[1 + 50sC]}$$

$$\omega_{-3dB} = \text{pole} = \frac{1}{50C} = 2\pi \times 10^{+9}$$

$$\therefore C = \frac{10^{-11}}{\pi} F = \frac{10}{\pi} pF = 3.183 \, pF$$

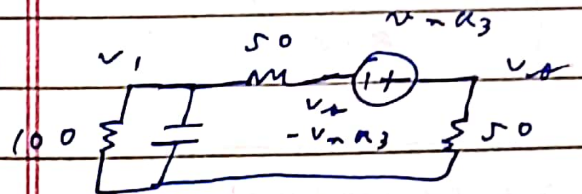
b) $$H(s) = \frac{0.25}{1+j100\pi f C} = \frac{0.25}{1+\frac{jf}{10^{9}}}$$

c) $$v_o = v_{nR_s} \times H(s)$$

$$\therefore S_{v_{no}} = \frac{S_{v_{nR_s}}}{16\left[1+(f\times10^{-9})^2\right]} = \frac{50kT}{4\left[1+(f\times10^{-9})^2\right]}$$

d) ∴ Same transfer function as R_s, and $R_1 = R_s$

$$\therefore S_{v_{no1}} = \frac{50kT}{4\left[1+(f\times10^{-9})^2\right]}$$

e) 

$$\frac{v_1 - (v_o - v_{nR_3})}{50} = \frac{v_o}{50}$$

$$\therefore v_1 = 2v_o - v_{nR_3}$$

$$\frac{-v_o}{50} = \frac{v_1}{100} + v_1 sC = (2v_o - v_{nR_3})\left(\frac{1+100sC}{100}\right)$$

$$v_{nR_3}(1+100sC) = 2v_o(2+100sC)$$

$$v_o = \frac{v_{nR_3}}{4}\left(\frac{1+100sC}{1+50sC}\right)$$

$$\therefore S_{v_{o3}} = \frac{kT\times50}{4}\left[\frac{1+(200\pi fC)^2}{1+(100\pi fC)^2}\right]$$

f) $$F = 1 + \frac{\frac{50 \times 7}{4}\left[1 + 1 + (2f \times 10^{-9})^2\right] \times \frac{1}{1 + (f \times 10^{-9})^2}}{\frac{50 \times 7}{4} \times \frac{1}{1 + (f \times 10^{-9})^2}}$$

$$F = 1 + 2 + 4f^2 \times 10^{-18} = 3 + 4f^2 \times 10^{-18}$$

$$NF = 10 \log_{10}\left[3 + 4f^2 \times 10^{-18}\right]$$

g) $f = 0.5\ GHz$:- $NF = 10 \log_{10}(3+1) = 6.02\ dB$

$f = 1\ GHz$:- $NF = 10 \log_{10}(3+4) = 8.45\ dB$

$f = 2\ GHz$:- $NF = 10 \log_{10}(3+16) = 12.79\ dB$

Q.3. a) $\omega_0 = 2\pi \times 10^9 = \frac{1}{\sqrt{LC}}$

- $Q_{inductor} = 25$ @ ω_0

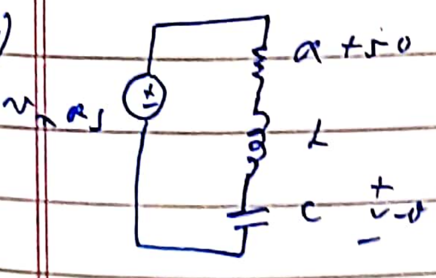
  $\therefore 25 = \frac{\omega_0 L}{R}$

- At ω_0, $|H(s)| = Q_{net} = 0.5 = \frac{\omega_0 L}{R + 50}$

- $\therefore \frac{25}{0.5} = \frac{R+50}{R} \Rightarrow R = \frac{50}{49}\,\Omega = 1.02\,\Omega$

- $25 = \frac{2\pi \times 10^9 \times L}{50/49} \Rightarrow L = 4.06\,nH$

- $C = \frac{1}{L \times 4\pi^2 \times 10^{18}} = 6.24\,pF$

b) 

$R_{net} = (R+50) + sL + \frac{1}{sC}$

$i = \frac{v_{n,R_s}}{R_{net}} \Rightarrow v_o = i \times \frac{1}{sC}$

$\therefore v_o = v_{n,R_s} \times \frac{1}{s^2LC + sC(R+50) + 1}$

$\therefore S_{v_{n,o}} = \frac{4KTR_s}{(1-\omega^2 LC)^2 + \omega^2 C^2 (R+50)^2}$

Date
Page

c) Same H(s) from a as from a_s :-

$$v_o = \frac{v_n a}{1-\omega^2 LC + j\omega C(a+50)}$$

$$\therefore S_{v_n o} = \frac{4kTa}{(1-\omega^2 LC)^2 + \omega^2 C^2 (a+50)^2}$$

d) $F = 1 + \frac{4kTa}{4kTa_s} = 1 + \frac{1}{49} = \frac{50}{49}$

$NF = 10\log_{10} F = 0.087\,dB$

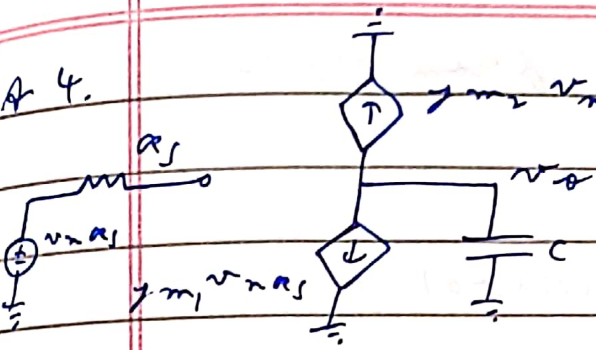
e) Q index doubles ⇒ a halves (for same L, ω_0)

$\therefore a_{new} = \frac{25}{49}$

$\therefore F_{new} = 1 + \frac{25}{50\times 49} = 1 + \frac{1}{98} = \frac{99}{98}$

$\therefore NF_{new} = 10\log_{10} F_{new} = 0.044\,dB$

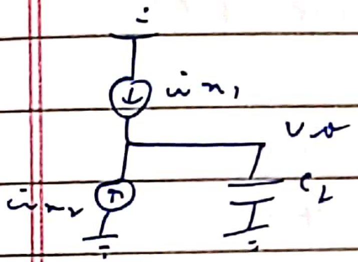
A 4.



$$\therefore v_o = -\frac{(g_{m_1} + g_{m_2}) v_{n R_s}}{sC}$$

$$\therefore S_{v_{n o}} = \left(\frac{g_{m_1} + g_{m_2}}{2\pi f C}\right)^2 \times 4kTR_s$$

•



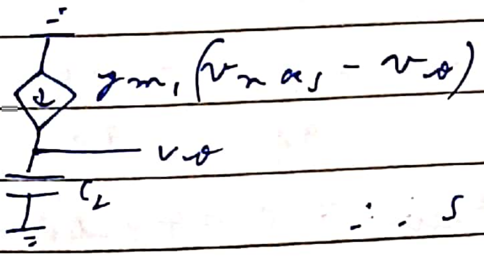
$$v_o \times sC_L = i_{n_1} + i_{n_2}$$

$$\therefore S_{v_o} = \frac{8}{3} \frac{kT(g_{m_1} + g_{m_2})}{(2\pi f C)^2}$$

$$\therefore F = 1 + \frac{\frac{8}{3} kT(g_{m_1} + g_{m_2}) \times (2\pi f C)^2}{(g_{m_1} + g_{m_2})^2 \times 4kTR_s \; (2\pi f C)^2}$$

$$F = 1 + \frac{2}{3R_s(g_{m_1} + g_{m_2})} \qquad \Big\{ NF = 10 \log_{10} F$$

A 5.

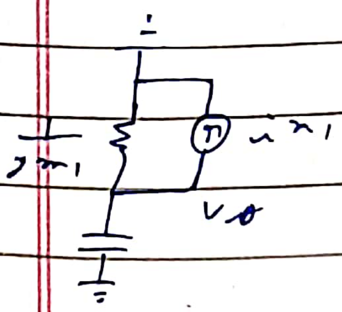


$$\therefore g_{m_1}(v_{n R_s} - v_o) = v_o sC$$

$$v_o(sC + g_{m_1}) = g_{m_1} v_{n R_s}$$

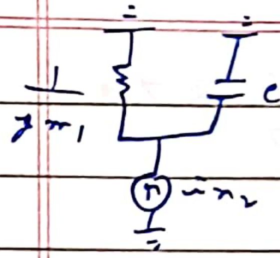
$$\therefore S_{v_{n o}} = \frac{g_{m_1}^2}{g_{m_1}^2 + (2\pi f C)^2} \times 4kTR_s$$

•



$$-i_{n_1} = v_o(g_{m_1} + sC)$$

$$\therefore S_{v_{n o_1}} = \frac{1}{g_{m_1}^2 + (2\pi f C)^2} \times \frac{8}{3} kT g_{m_1}$$



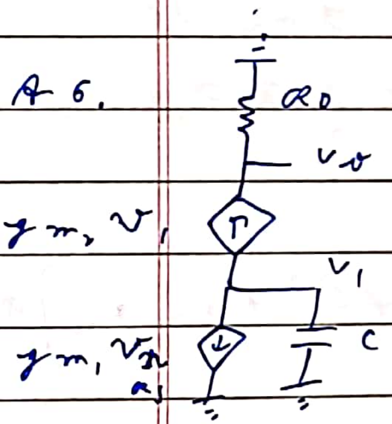
$$i_{n2} = v_o (g_{m1} + sC)$$

$$\therefore S_{v_{no2}} = \frac{1}{g_{m1}^2 + (2\pi f C)^2} \times \frac{8}{3} kT g_{m2}$$

$$F = 1 + \frac{\frac{8}{3} kT (g_{m1} + g_{m2})}{4kT R_s g_{m1}^2}$$

$$F = 1 + \frac{2}{3}\left(\frac{g_{m1} + g_{m2}}{g_{m1}^2 R_s}\right) \quad \Bigg| \quad NF = 10 \log_{10} F$$

A 6.



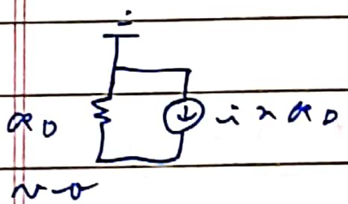
$$g_{m2} v_1 + g_{m1} v_{nR_s} + v_1 sC = 0$$

$$\therefore v_1 = \frac{-g_{m1}}{g_{m2} + sC} v_{nR_s}$$

$$v_o = g_{m2} R_D v_1$$

$$\therefore S_{v_{no}} = \frac{(g_{m1} g_{m2} R_D)^2}{g_{m2}^2 + (2\pi f C)^2} 4kT R_s$$

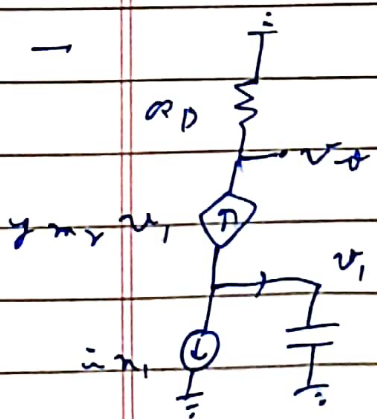
—



$$v_o = i_{nR_D} \times R_D$$

$$\therefore S_{v_{no1}} = 4kT R_D$$

—

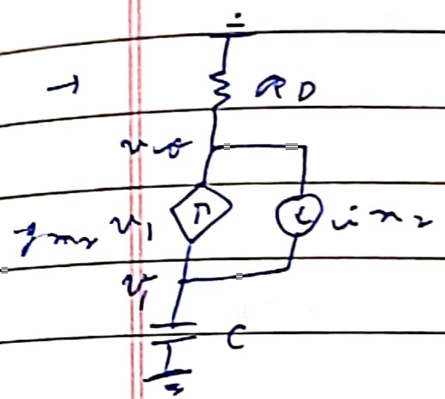


$$g_{m2} v_1 + i_{n1} + v_1 sC = 0$$

$$v_1 = \frac{-i_{n1}}{g_{m2} + sC}$$

$$v_o = \frac{-g_{m2} R_D i_{n1}}{g_{m2} + sC}$$

$$SV_{n,o2} = \frac{(g_{m2}R_D)^2}{g_{m2}^2 + (2\pi f C)^2} \times \frac{8}{3}kTg_{m1}$$



$$g_{m1}v_1 + v_1 sC = i_{n2}$$

$$v_1 = \frac{i_{n2}}{g_{m2} + sC}$$

$$\frac{v_o}{R_D} = -v_1 sC \Rightarrow v_o = \frac{-sCR_D\, i_{n2}}{g_{m2} + sC}$$

$$\therefore SV_{n,o3} = \frac{(2\pi f C R_D)^2}{g_{m2}^2 + (2\pi f C)^2} \times \frac{8}{3}kTg_{m2}$$

$$F = 1 + \frac{4kTR_D + \dfrac{\frac{8}{3}kTg_{m2}\left[(2\pi f C)^2 + g_{m1}g_{m2}\right]R_D^2}{g_{m2}^2 + (2\pi f C)^2}}{\dfrac{(g_{m1}g_{m2}R_D)^2 \times 4kTR_S}{g_{m2}^2 + (2\pi f C)^2}}$$

$$F = 1 + \frac{g_{m2}^2\left(1 + \frac{2}{3}R_D g_{m1}\right) + (2\pi f C)^2\left(1 + \frac{2}{3}R_D g_{m2}\right)}{(g_{m1}g_{m2})^2 R_D R_S}$$

$$NF = 10\log_{10} F$$